MA1521 CALCULUS FOR COMPUTING
National University of Singapore

Exercise Set M (Midterm Revision)
Name: _____
Date: _____

Vectors, Space Curves, and Partial Derivatives

Thomas' Calculus: Early Transcendentals, 15/e • Sections 11.1–11.5, 12.1–12.3, and Chapter 13 complete
MIT 18.02 Multivariable Calculus • Lectures 1–13

HOW TO USE THIS SET
The set is arranged in five parts. Parts A and B rebuild the geometric machinery (§11.1–11.5, §12.1–12.3); Parts C and D are the heart of the midterm and cover Chapter 13 in full, from level curves through Lagrange multipliers; Part E is a timed simulation assembled from all of the above.
Every problem is followed by a ruled panel sized to the work it should take. Do the problem *inside* the panel — if your work spills out of the box, you are almost certainly computing something the problem did not ask for. Answers to all problems begin on the last pages; consult them only after a complete attempt.
Problems are drawn from the exercise sets of Thomas' Calculus 15/e, from the MIT 18.02 problem sets used in Lecture Group 1, and from the tutorial questions of Lecture Group 2. The source of each part is recorded in its banner.

Distribution of Problems

Part	Topic	Thomas sections	Problems
A	Vectors and the Geometry of Space	11.1–11.5	1–9
B	Vector-Valued Functions and Motion	12.1–12.3	10–15
C	Partial Derivatives, Chain Rule, Gradient	13.1–13.6	16–38
D	Extreme Values and Lagrange Multipliers	13.7–13.9	39–52
E	Midterm Simulation (90 minutes)	11–13	53–58

Results You Are Expected to Know Cold

THEOREM 1 — Angle Between Two Vectors
The angle θ between two nonzero vectors $\mathbf{u} = \langle u_1, u_2, u_3\rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ is

$$\theta = \cos^{-1}\!\left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}|\,|\mathbf{v}|}\right), \qquad \mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3.$$

$$\operatorname{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\right)\mathbf{v}, \qquad |\mathbf{u}\times\mathbf{v}| = |\mathbf{u}|\,|\mathbf{v}|\sin\theta, \qquad |(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}| = |\det[\mathbf{u}\ \mathbf{v}\ \mathbf{w}]|.$$

DEFINITION — Arc Length Along a Space Curve
For a smooth curve $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, $a \le t \le b$,

$$L = \int_a^b |\mathbf{v}(t)|\,dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\,dt, \qquad \mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|}.$$

THEOREM — The Chain Rule, the Gradient, and the Second Derivative Test

If $w = f(x, y, z)$ is differentiable and x, y, z are differentiable functions of t, then $\dfrac{dw}{dt} = f_x \dfrac{dx}{dt} + f_y \dfrac{dy}{dt} + f_z \dfrac{dz}{dt}$, and $df = f_x\,dx + f_y\,dy + f_z\,dz$.

The directional derivative of f at P_0 in the direction of a *unit* vector $\mathbf{u}$ is $D_{\mathbf{u}} f = \nabla f|_{P_0} \cdot \mathbf{u}$; it is largest ($= |\nabla f|$) in the direction of ∇f and zero along the level curve through P_0.

At a critical point (a, b) of f, set $D = f_{xx} f_{yy} - f_{xy}^2$ evaluated at (a, b). Then f has a local maximum if $D > 0$ and $f_{xx} < 0$; a local minimum if $D > 0$ and $f_{xx} > 0$; a saddle point if $D < 0$; and the test is inconclusive if $D = 0$.

Lagrange multipliers. To find the extreme values of f subject to $g = 0$, solve $\nabla f = \lambda \nabla g$ together with $g = 0$, then compare the values of f at the solutions.

PART A

9 problems | 25 points

Vectors and the Geometry of Space

Thomas 11.1–11.5 • MIT 18.02 Lectures 1–5

Vectors, Lengths, and Directions

1. **Unit vectors and displacement.**

(a) For what value or values of c will $\frac{1}{2}\mathbf{i} - c\mathbf{j} + c\mathbf{k}$ be a unit vector?

(b) A vector $\mathbf{A}$ has magnitude 6 and direction $(\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})/3$. If its tail is at $(-2, 0, 1)$, where is its head?

[MIT 18.02, Problem Set 1]

2. Let P and Q be two points in space, and let X be the midpoint of the line segment PQ. Let O be an arbitrary fixed point.

(a) Show that, as vectors, $\overrightarrow{OX} = \frac{1}{2}(\overrightarrow{OP} + \overrightarrow{OQ})$.

(b) With the notation of part (a), assume instead that X divides the segment PQ in the ratio $r : s$, where $r + s = 1$. Derive an expression for $\overrightarrow{OX}$ in terms of $\overrightarrow{OP}$ and $\overrightarrow{OQ}$.

[MIT 18.02, Problem Set 1]

The Dot Product, Projections, and Angles

3. **Angles between vectors.**

(a) Find the angle between $\mathbf{i} - \mathbf{k}$ and $4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$, and the angle between $\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + \mathbf{k}$.

(b) Tell for what values of c the vectors $c\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ will (i) be orthogonal; (ii) form an acute angle.

[MIT 18.02, Problem Set 1]

4. Consider the vectors $\mathbf{A} = 6\mathbf{j} + 6\mathbf{k}$ and $\mathbf{B} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$. Write $\mathbf{A} = \mathbf{C} + \mathbf{D}$ as a sum of two vectors such that $\mathbf{C}$ is parallel to $\mathbf{B}$ and $\mathbf{D}$ is perpendicular to $\mathbf{B}$.

Suggestion. First write $\mathbf{C} = \alpha\mathbf{B}$*, so that* $\mathbf{A} = \alpha\mathbf{B} + \mathbf{D}$ *for some real number* α*. Determine* α *using the dot product, then proceed to get* $\mathbf{C}$ *and* $\mathbf{D}$*.*

[MA1521 L2, Q04.4]

Determinants and the Cross Product

5. Determinants and the cross product.

(a) Calculate

$$\begin{vmatrix} -1 & 0 & 4 \\ 1 & 2 & 2 \\ 3 & -2 & -1 \end{vmatrix}$$

using the Laplace expansion by the cofactors of (i) the first row; (ii) the first column.

(b) Find $\mathbf{A} \times \mathbf{B}$ if (i) $\mathbf{A} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{B} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$; (ii) $\mathbf{A} = 2\mathbf{i} - 3\mathbf{k}$, $\mathbf{B} = \mathbf{i} + \mathbf{j} - \mathbf{k}$.

[MIT 18.02, Problem Set 1]

6. Consider the points $A = (5, 2, 0)$, $B = (2, 6, 1)$, and $C = (2, 4, 7)$ in space.

(a) Find the area of the parallelogram with two adjacent sides given by the line segments AB and AC. Hence find the area of triangle ABC.

(b) Find the volume of the parallelepiped spanned by $\mathbf{i} + 3\mathbf{k}$, $2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, and $5\mathbf{i} + 4\mathbf{k}$.

(c) Suppose A, B, C, D are four points in space with $\det(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}) = 0$. Must the four points be coplanar? Give a reason.

[MA1521 L2, Q04.8–Q04.10]

Lines and Planes in Space

7. Find equations for the following planes.

(a) Through $(2, 0, -1)$ and perpendicular to $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$.

(b) Through the origin, $(1, 1, 0)$, and $(2, -1, 3)$.

(c) Through the points on the x-, y-, and z-axes where $x = a$, $y = b$, $z = c$ respectively (all nonzero).

(d) Through $(1, 0, 1)$ and $(0, 1, 1)$ and parallel to $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

[MIT 18.02, Problem Set 1]

8. Find, in parametric form, equations for

(a) the line through $(1, 0, -1)$ and parallel to $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$;

(b) the line through $(2, -1, -1)$ and perpendicular to the plane $x - y + 2z = 3$;

(c) all lines passing through $(1, 1, 1)$ and lying in the plane $x + 2y - z = 2$;

(d) the line through $P(4, 2, 1)$ and perpendicular to both of the lines

$$\ell_1 : x = 3t + 2, \quad y = -4t - 1, \quad z = 4t - 9, \qquad \ell_2 : x = 3t - 4, \quad y = -t + 6, \quad z = 5t.$$

[MIT 18.02, Problem Set 1; MA1521 L2, Q05.2]

9. Lines meeting planes.

(a) Where does the line through $(0, 1, 2)$ and $(2, 0, 3)$ intersect the plane $x + 4y + z = 4$?

(b) The line passing through $(1, 1, -1)$ and perpendicular to the plane $x + 2y - z = 3$ intersects the plane $2x - y + z = 1$ at what point?

(c) Find an equation of the plane that contains the point $(-1, 2, 1)$ and the line of intersection of the planes $x + y - z = 2$ and $2x - y + 3z = 1$.

[MIT 18.02, Problem Set 1; MA1521 L2, Q05.5]

PART B

6 problems | 20 points

Vector-Valued Functions and Motion in Space

Thomas 12.1–12.3 • MIT 18.02 Lectures 5–6

Curves in Space and Their Tangents

10. For each of the following vector functions of time, calculate the velocity, the speed, the unit tangent vector (in the direction of the velocity), and the acceleration.

(a) $\mathbf{r} = e^t\mathbf{i} + e^{-t}\mathbf{j}$

(b) $\mathbf{r} = t^2\mathbf{i} + t^3\mathbf{j}$

(c) $\mathbf{r} = (1 - 2t^2)\mathbf{i} + t^2\mathbf{j} + (-2 + 2t^2)\mathbf{k}$

[MIT 18.02, Problem Set 2]

11. Consider the vector function $\mathbf{v}(t) = (3t^2 + 2t)\mathbf{i} + \pi\sin(\pi t)\mathbf{j} + e^t\mathbf{k}$.

(a) Evaluate $\mathbf{v}'(t)$, and find $\mathbf{v}'(2)$.

(b) Evaluate the indefinite integral $\int \mathbf{v}(t)\,dt$.

(c) Evaluate the definite integral $\int_0^1 \mathbf{v}(t)\,dt$.

[MA1521 L2, Q05.10]

12. Consider the curve in $\mathbb{R}^3$ given by

$$\mathbf{r}(t) = (\ln t)\,\mathbf{i} + \frac{t-1}{t+3}\,\mathbf{j} + (t \ln t)\,\mathbf{k}.$$

Find parametric equations of the tangent line to the curve at the point where $t = 1$.

[MA1521 L2, Q05.11]

Integrals of Vector Functions; Motion

13. A particle moves in $\mathbb{R}^3$ so that its velocity at time t is

$$\mathbf{v}(t) = (6t^2 + 4t)\,\mathbf{i} + (8t^3 + 2t)\,\mathbf{j} + \frac{1}{1+t^2}\,\mathbf{k}.$$

If the position of the particle at time $t = 0$ is $\mathbf{i} + 2\mathbf{j} + \pi\mathbf{k}$, find the position of the particle at time $t = 1$.

[MA1521 L2, Q05.12]

14. Two particles move in $\mathbb{R}^3$ so that their positions at time t are

$$\mathbf{r}_1(t) = t\,\mathbf{i} + t^2\,\mathbf{j} + t^3\,\mathbf{k}, \qquad \mathbf{r}_2(t) = (1+2t)\,\mathbf{i} + (1+6t)\,\mathbf{j} + (1+14t)\,\mathbf{k}.$$

(a) Do the particles collide? Justify your answer.

(b) Do their paths intersect? Justify your answer.

[MA1521 L2, Q05.13]

Arc Length in Space

15. **Lengths of curves in space.**

(a) Find the unit tangent vector of $\mathbf{r}(t) = (2\cos t)\mathbf{i} + (2\sin t)\mathbf{j} + \sqrt{5}\,t\,\mathbf{k}$, and the length of the portion $0 \le t \le \pi$.

(b) Find the arc length of the curve

$$x = 2t - 2\sin t, \qquad y = 2 - \sqrt{2}\cos t, \qquad z = \sqrt{2}\cos t, \qquad 0 \le t \le 4\pi.$$

(c) Find the length of the curve $\mathbf{r}(t) = 2t\,\mathbf{i} + 2t\,\mathbf{j} + (1-t^2)\,\mathbf{k}$ from $(0,0,1)$ to $(2,2,0)$.

[Thomas 12.3, Exercises 1 and 15; MA1521 L2, Q05.14]

PART C

23 problems | 60 points

Partial Derivatives, the Chain Rule, and the Gradient

Thomas 13.1–13.6 • MIT 18.02 Lectures 8–12

Functions of Several Variables; Limits and Continuity

16. Values, domains, and level sets.

(a) Let $f(x,y,z)=\dfrac{x-y}{y^2+z^2}$. Evaluate $f(3,-1,2)$ and $f\left(0,-\tfrac{1}{3},0\right)$.

(b) Find and sketch the natural domain of $f(x,y)=\dfrac{1}{\ln\left(9-x^2-y^2\right)}$.

(c) Describe geometrically the level sets of $f(x,y,z)=\sqrt{16-x^2-y^2-z^2}$.

[MA1521 L2, Q06.1, Q06.2, Q06.4]

17. **Level curves and graphs.**

(a) Sketch five level curves for each of the following functions, and sketch the portion of the graph lying in the first octant:

$$\text{(i) } 1-x-y, \qquad \text{(ii) } \sqrt{x^2+y^2}, \qquad \text{(iii) } x^2+y^2, \qquad \text{(iv) } 1-x^2-y^2.$$

(b) Sketch the level curves $f(x,y)=k$ for $k=-2,0,1,2$, where $f(x,y)=\dfrac{x+y}{x-y}$.

[MIT 18.02, Problem Set 3; MA1521 L2, Q06.3]

18. Evaluate each limit, or show that it does not exist.

(a) $\displaystyle\lim_{(x,y)\to(0,\pi/4)} \sec x \tan(x+y)$

(b) $\displaystyle\lim_{(x,y)\to(2,0)} \frac{\sqrt{2x-y}-2}{2x-y-4}$ *(The identity $a^2-b^2=(a+b)(a-b)$ is useful.)*

(c) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{y}{x^2}$

(d) Let $f(x,y)=\sqrt{x^2+y^2}$ for $(x,y)\neq(0,0)$ and $f(0,0)=1$. Find $\displaystyle\lim_{(x,y)\to(0,0)} f(x,y)$, and decide whether f is continuous at $(0,0)$.

[MA1521 L2, Q06.5, Q06.6, Q07.1]

Partial Derivatives

19. Calculate the first partial derivatives of each of the following functions.

(a) $4x^3 - 3x^2y^2 + 2x$

(b) $\sin(3x + 2y)$

(c) $y\ln(4x + 3y)$; find also the values of the first partials at $(2, 1)$.

(d) $x^2z - 2yz^3$

(e) $y^2z + x^3yz^3$; find also the values of the first partials at $(1, 2, 4)$.

[MIT 18.02, Problem Set 3; MA1521 L2, Q07.2]

20. **Mixed second partials.**

(a) Verify that $f_{xy} = f_{yx}$ for (i) $x^m y^n$ (m, n positive integers); (ii) $\dfrac{x}{1+y}$; (iii) $\cos(x^2 + y)$.

(b) Using $f_{xy} = f_{yx}$, tell for what value of the constant a there exists a function $f(x, y)$ with $f_x = axy + 3y^2$ and $f_y = x^2 + 6xy$; then find such a function by inspection.

(c) Suppose $f(x, y)$ satisfies $f_x = 3x^2 + Axy$ and $f_y = 3x^2 + 6y^2$, where A is a constant. Using Clairaut's Theorem, determine A, and then find such a function $f(x, y)$. *(Such a function is not unique.)*

[MIT 18.02, Problem Set 3; MA1521 L2, Q07.6]

21. Laplace's equation and implicit partials.

(a) Show that each of the following functions $w = f(x, y)$ satisfies the two-dimensional Laplace equation $w_{xx} + w_{yy} = 0$: (i) $w = e^{ax} \sin ay$ (a constant); (ii) $w = \ln(x^2 + y^2)$.

(b) Using implicit differentiation, find $\partial z/\partial x$ and $\partial z/\partial y$ if

$$x^3 + 2y^3 + 6xz^3 - 2 = 59z.$$

Here z is regarded as a function of the independent variables x and y, defined implicitly by the equation.

[MIT 18.02, Problem Set 3; MA1521 L2, Q07.3]

Tangent Planes and Normal Lines

22. Give an equation of the tangent plane to each of the following surfaces at the point indicated.

(a) $z = xy^2$ at $(1, 1, 1)$

(b) $w = y^2/x$ at $(1, 2, 4)$

(c) the graph of $f(x, y) = x^2y + 2xy$ at the point where $(x, y) = (1, 2)$

(d) the surface $yz = x^2$ at $(4, 2, 8)$ *(Suggestion: first rewrite the surface as the graph of a function of two variables.)*

[MIT 18.02, Problem Set 3; MA1521 L2, Q07.8, Q07.9]

23. Consider the surface in xyz-space defined by

$$\cos(\pi x) - x^2 y + e^{xz} + yz = 4.$$

Find equations for the tangent plane and the normal line to the surface at the point $(0, 1, 2)$.

[Thomas 13.6, Exercise 5; MA1521 L2, Q10.1]

24. **Tangent planes to a cone.**

(a) Find an equation of the tangent plane to the cone $z = \sqrt{x^2 + y^2}$ at a point $P_0 : (x_0, y_0, z_0)$ on the cone, with $z_0 > 0$.

(b) Write parametric equations for the ray from the origin passing through P_0, and use them to show that the ray lies on both the cone and the tangent plane at P_0.

[MIT 18.02, Problem Set 3]

Differentials

25. Find the differential (dw or dz). Make the answer look as neat as possible.

(a) $w = \ln(xyz)$

(b) $w = xy^2z^3$

(c) $z = \dfrac{x - y}{x + y}$

[MIT 18.02, Problem Set 4; MA1521 L2, Q10.5]

26. The following equations define w implicitly as a function of the other variables. Find dw in terms of the other differentials by taking the differential of both sides and solving algebraically for dw.

(a) $z^2 + xyw + yw^3 = 0$

(b) $u^2 + 2v^2 + 3w^2 = 10$

[MIT 18.02, Problem Set 4; MA1521 L2, Q10.6]

The Chain Rule

27. In each of the following, find dw/dt in two distinct ways: (i) use the Chain Rule, then express the answer in terms of t; (ii) express the composite function in terms of t and differentiate.

(a) $w = xyz; \quad x = t,\ y = t^2,\ z = t^3$

(b) $w = x^2 - y^2; \quad x = \cos t,\ y = \sin t$

(c) $w = \ln\left(u^2 + v^2\right); \quad u = 2\cos t,\ v = 2\sin t$

[MIT 18.02, Problem Set 4]

28. Consider the function $w = 2ye^x - \ln z$, where $x = \ln\left(t^2 + 1\right)$, $y = \tan^{-1} t$, and $z = e^t$.

(a) Find dw/dt using the Chain Rule, expressing the answer in terms of t.

(b) Find dw/dt by first expressing w in terms of t, and check that the two answers agree.

[MA1521 L2, Q09.2]

29. Suppose the temperature at any point in xyz-space is $T(x, y, z) = e^x - yz + zx$, measured in ˚C, where x, y, z are measured in cm. If the position of an insect at time t seconds is $x = t$, $y = 2\left(1 + t^2\right)$, $z = 3 + t$, find the rate of change of temperature experienced by the insect at time $t = 0$. Give the answer in ˚C per second.

[MA1521 L2, Q09.3]

30. Let $w = f(x, y)$, where all first- and second-order partial derivatives of f are continuous, and change to polar coordinates $x = r\cos\theta$, $y = r\sin\theta$.

(a) Show that $(w_x)^2 + (w_y)^2 = (w_r)^2 + \frac{1}{r^2}(w_\theta)^2$.

(b) Show that $\dfrac{\partial^2 w}{\partial r^2} = f_{xx}\cos^2\theta + 2f_{xy}\cos\theta\sin\theta + f_{yy}\sin^2\theta$. *(First find $\partial w/\partial r$ in terms of f_x, f_y, and θ. Clairaut's Theorem may be used freely.)*

[MIT 18.02, Problem Set 4; MA1521 L2, Q09.6]

31. Let $w = f(x, y)$ and make the change of variables $x = u^2 - v^2$, $y = 2uv$. Show that

$$(w_u)^2 + (w_v)^2 = 4\big(u^2 + v^2\big)\Big[(w_x)^2 + (w_y)^2\Big].$$

[MIT 18.02, Problem Set 4]

32. **Partial differential equations from the Chain Rule.**

(a) Let $w = f(y/x)$; that is, w is the composite of $w = f(u)$ and $u = y/x$. Show that w satisfies the partial differential equation $x\,\dfrac{\partial w}{\partial x} + y\,\dfrac{\partial w}{\partial y} = 0$.

(b) Let $w = f\big(x^2 - y^2\big)$; show that $y\,\dfrac{\partial w}{\partial x} + x\,\dfrac{\partial w}{\partial y} = 0$.

(c) Let $w = f(ax + by)$; show that $b\,\dfrac{\partial w}{\partial x} - a\,\dfrac{\partial w}{\partial y} = 0$.

[MIT 18.02, Problem Set 4]

Directional Derivatives and Gradient Vectors

33. In each of the following, a function f, a point P, and a vector $\mathbf{A}$ are given. Calculate the gradient of f at P, and the directional derivative $D_{\mathbf{u}}f$ at P in the direction $\mathbf{u}$ of the given vector $\mathbf{A}$.

(a) $f(x,y) = 2xy - 3y^2$; $P(5,5)$, $\mathbf{A} = 4\mathbf{i} + 3\mathbf{j}$

(b) $f(x,y,z) = xy + yz + zx$; $P(1,-1,2)$, $\mathbf{A} = 3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$

(c) $f(x,y,z) = x^2 + 2y^2 - 3z^2$; $P(1,1,1)$, $\mathbf{A} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

(d) $f(u,v,w) = (u + 2v + 3w)^2$; $P(1,-1,1)$, $\mathbf{A} = -2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$

[Thomas 13.5, Exercises 11, 15, 16; MIT 18.02, Problem Set 4]

34. Consider the function $f(x,y) = 2ye^{-x} + 1$ and the point $P(0,-2)$.

(a) Find the directional derivative of f at P in the direction of each of the vectors (i) $\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$; (ii) $12\mathbf{i} + 5\mathbf{j}$.

(b) Find the direction that gives the largest possible directional derivative of f at P, and the value of that derivative.

(c) Find the direction along which f decreases most rapidly at P.

(d) Find the directions along which f exhibits no change at P.

[MA1521 L2, Q09.10]

35. Let a be a positive number and consider $f(x,y) = \big(x^2 + a\big)e^{2x+3y^2}$. It is given that the directional derivative of f at the point $(-6,2)$ in the direction of the vector $\mathbf{i} + 2\mathbf{j}$ is $216\sqrt{5}$. Find the value of a.

[MA1521 L2, Q09.9]

36. Let C be the curve of intersection of the two surfaces $x^3 + 2xy + yz = 7$ and $3x^2 - yz = 1$. Find parametric equations of the tangent line to C at $(1,2,1)$.

Suggestion. For each of the two surfaces, first find a normal vector at the point $(1,2,1)$.

[Thomas 13.6, Exercises 15–20; MA1521 L2, Q10.2]

Linearization and Quadratic Approximation

37. Linearization and linear approximation.

(a) Find the linearization of $f(x, y) = e^{2x+y}$ at $(0, 0)$ and at $(1, 2)$. Give each answer in the form $ax + by + c$.

(b) Consider $f(x, y, z) = xz + \sin(xyz)$. Using linear approximation, estimate the change in the value of f when (x, y, z) changes from $\left(\frac{1}{2}, \pi, \frac{1}{3}\right)$ to $\left(\frac{1}{2} + 0.03, \pi + 0.04, \frac{1}{3} - 0.02\right)$.

[MA1521 L2, Q10.4, Q10.7]

38. Consider the function $f(x, y) = e^{x+4y} \cos x$.

(a) Find the second-degree Taylor polynomial of f at $(0, 0)$.

(b) Using part (a), find an approximate value of $f(0.2, 0.1)$.

[Thomas 13.9; MA1521 L2, Q10.9]

PART D

14 problems | 60 points

Extreme Values, Saddle Points, and Lagrange Multipliers

Thomas 13.7–13.9 • MIT 18.02 Lectures 9, 10, 13

Finding Local Extrema

39. Find all the local maxima, local minima, and saddle points of the following functions, and give the corresponding values of the function.

(a) $f(x, y) = x^2 + xy + y^2 + 3x - 3y + 4$

(b) $f(x, y) = x^3 - y^3 - 2xy + 6$

(c) $f(x, y) = 4xy - x^4 - y^4$

(d) $f(x, y) = 2\ln x + \ln y - 4x - y \quad (x > 0,\, y > 0)$

[Thomas 13.7, Exercises 1, 13, 19, 29]

40. Determine all the critical points of

$$f(x, y) = 2xy - x^2y - xy^2,$$

and classify each as a local maximum point, a local minimum point, or a saddle point of f. Evaluate also the local maximum and local minimum values of f, if any.

[MA1521 L2, Q08.2]

41. Determine all the critical points of

$$f(x, y) = \ln\left(xy^2\right) - xy - 2y$$

over the region $x > 0$, $y > 0$ in the xy-plane, and classify each one. Evaluate also the local maximum and local minimum values of f, if any.

[MA1521 L2, Q08.3]

Finding Absolute Extrema

42. Find the maximum and minimum points of

$$f(x, y) = x^2 - 2xy + 2y^2 - 2y$$

over the rectangular region $R : -1 \leq x \leq 2,\ 0 \leq y \leq 2$ in the xy-plane. Hence find the absolute maximum and absolute minimum values of f over R.

[MIT 18.02, Problem Set 3; MA1521 L2, Q08.6]

43. **Constrained minimum distance.**

(a) Consider the surface $y^2 - xz = 4$ in xyz-space. Find the point or points on the surface closest to the origin. *(Suggestion: first formulate the question as a problem of minimizing a function of the variables x and z.)*

(b) Find the minimum distance from the point $(2, -1, 1)$ to the plane $x + y - z = 2$.

[Thomas 13.7, Exercise 52; MA1521 L2, Q08.4]

44. The method of least squares.

(a) Using the method of least squares, find the line $y = ax + b$ that best fits the three data points $(-1, 1)$, $(1, 2)$, $(3, 2)$.

(b) Using the method of least squares, find the quadratic polynomial of the form $y = ax^2 + bx$ that best fits the three data points $(-1, 1)$, $(1, 1)$, $(2, 0)$.

[MIT 18.02, Problem Set 3; MA1521 L2, Q08.7, Q08.8]

45. A rectangular box with its top left open has its five faces made from different materials. The front face and the back face are made from material costing 10 dollars per square metre; the two lateral faces from material costing 5 dollars per square metre; and the bottom face from material costing 30 dollars per square metre. Find the minimum material cost of the box if its volume is 6 cubic metres, and give the dimensions (length, width, and height) of such a box.

Do this problem twice: first by eliminating a variable and using the Second Derivative Test, and then again by the method of Lagrange multipliers. Compare the work involved.

[MIT 18.02, Problem Set 3; MA1521 L2, Q08.5 and Q11.4]

The Method of Lagrange Multipliers

46. Extrema on a circle.

(a) Use the method of Lagrange multipliers to find the maximum value of $f(x, y) = xy$ subject to the constraint $(x-2)^2 + y^2 = 4$.

(b) Use the method of Lagrange multipliers to find the maximum value of $f(x, y) = xy$ subject to the constraints $(x-2)^2 + y^2 = 4$ *and* $x \leq 2$. *(This is part (a) with one extra constraint, so the work there is useful.)*

[MA1521 L2, Q11.1, Q11.2]

47. Nearest and farthest points.

(a) Find the point or points on the sphere $x^2 + y^2 + z^2 = 1$ furthest from the point $(2, 1, 2)$.

(b) Find the point on the plane $x + 2y + 3z = 13$ closest to the point $(1, 1, 1)$.

[Thomas 13.8, Exercise 17; MA1521 L2, Q11.3]

48. A rectangular box is placed in the first octant so that one corner Q is at the origin and the three sides adjacent to Q lie in the coordinate planes. The corner P diagonally opposite to Q lies on the ellipsoid

$$\frac{x^2}{3} + \frac{y^2}{9} + \frac{z^2}{4} = 1.$$

Use the method of Lagrange multipliers to determine the largest possible volume of the box, and say how you know that your answer gives a maximum.

[MIT 18.02, Problem Set 5; MA1521 L2, Q12.1]

49. Let $f(x, y) = x^2 - y^2$ be defined on the disk $D = \{(x, y) : x^2 + y^2 \leq 1\}$.

(a) Find the critical points of f and classify them.

(b) Find the maximum and minimum values of f on D. *(Lagrange multipliers are used in part (b).)*

[MA1521 L2, Q12.2]

50. Extrema on a curve of intersection.

(a) Find the extreme values of $f(x, y, z) = xy + z^2$ on the circle in which the plane $y - x = 0$ intersects the sphere $x^2 + y^2 + z^2 = 4$.

(b) Find the maximum value of $w = xyz$ on the line of intersection of the two planes $x + y + z = 40$ and $x + y - z = 0$. Give a geometric argument supporting the claim that you have found a maximum and not a minimum.

[Thomas 13.8, Exercises 42, 43]

51. Three applications.

(a) The temperature at a point (x, y) on a metal plate is $T(x, y) = 4x^2 - 4xy + y^2$. An ant on the plate walks around the circle of radius 5 centred at the origin. What are the highest and lowest temperatures the ant encounters?

(b) Find the dimensions of the closed right circular cylindrical can of smallest surface area whose volume is 16π cm^3.

(c) Suppose the utility of amounts x and y of two capital goods is $U(x, y) = xy + 2x$, and that the budget constraint simplifies to $2x + y = 30$. Find the maximum value of U and the corresponding values of x and y.

[Thomas 13.8, Exercises 9, 15, 33]

52. The condition $\nabla f = \lambda \nabla g$ is not sufficient. Even though $\nabla f = \lambda \nabla g$ is a necessary condition for an extreme value of $f(x, y)$ subject to $g(x, y) = 0$ and $\nabla g \neq \mathbf{0}$, it does not by itself guarantee that an extreme value exists. Try using Lagrange multipliers to find a maximum value of $f(x, y) = x + y$ subject to $xy = 16$. The method identifies $(4, 4)$ and $(-4, -4)$ as candidates. Explain carefully why $x + y$ nevertheless has *no* maximum value on the hyperbola $xy = 16$, and say which of the two candidates is a genuine local extremum and of what kind.

[Thomas 13.8, Exercise 45]

PART E

6 problems | 100 points

Midterm Simulation

Thomas 11–13 • 90 minutes, no calculator, closed book

INSTRUCTIONS
Work the six problems below in one uninterrupted sitting of **90 minutes**. Show all reasoning; an unsupported answer earns no credit. Leave answers in exact form ($\sqrt{2}$, $\ln 3$, $\pi/4$, and so on). The point values sum to 100.

53. [15 points]

Let $A = (1, 0, 2)$, $B = (2, 1, 0)$, and $C = (0, 3, 1)$.

(a) Find the area of the triangle ABC.

(b) Find an equation of the plane containing A, B, and C.

(c) Find the distance from the origin to that plane.

54. [13 points]

Consider the plane curve

$$\mathbf{r}(t) = (\cos t + t\sin t)\,\mathbf{i} + (\sin t - t\cos t)\,\mathbf{j}, \qquad t \geq 0.$$

(a) Find $\mathbf{v}(t)$ and the speed $|\mathbf{v}(t)|$, simplifying as far as possible.

(b) Find the arc length parameter $s(t)$ measured from the point where $t = 0$, and hence the length of the portion of the curve with $0 \leq t \leq \pi/2$.

(c) Find the unit tangent vector $\mathbf{T}(t)$ for $t > 0$.

55. [18 points]

Let $f(x, y, z) = x^2y + yz^2$ and let $P = (1, 2, -1)$.

(a) Find ∇f at P.

(b) Find the directional derivative of f at P in the direction of $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

(c) In what direction does f increase most rapidly at P, and what is that rate of increase?

(d) A particle moves so that its position at time t is $\mathbf{r}(t) = t\,\mathbf{i} + (t+1)\,\mathbf{j} + \big(t^2 - 2\big)\mathbf{k}$. Verify that the particle is at P when $t = 1$, and find the rate of change of f experienced by the particle at that instant.

56. [16 points]

Consider the ellipsoid $x^2 + 2y^2 + 3z^2 = 21$ and the point $P_0(1, 2, 2)$ on it.

(a) Find an equation of the tangent plane and parametric equations of the normal line to the ellipsoid at P_0.

(b) Regarding z as a function of x and y near P_0, use implicit differentiation to find $\partial z/\partial x$ and $\partial z/\partial y$ at P_0, and check that your answers agree with part (a).

(c) Find all points on the ellipsoid at which the tangent plane is parallel to the plane $x + 4y + 6z = 0$.

57. [18 points]

Find and classify all the critical points of

$$f(x, y) = x^3 + 3xy^2 - 15x - 12y,$$

using the Second Derivative Test. State the local maximum and local minimum values of f.

58. [20 points]

(a) Use the method of Lagrange multipliers to find the maximum and minimum values of $f(x, y, z) = x - 2y + 5z$ on the sphere $x^2 + y^2 + z^2 = 30$.

(b) Find the dimensions of the closed rectangular box of maximum volume that can be inscribed in the sphere $x^2 + y^2 + z^2 = 4$, and state that maximum volume.

Answers

Consult these only after a complete attempt. "Show that" problems are answered with the key step rather than the full argument.

Part A

1. (a) $c = \pm\sqrt{6}/4$. (b) $(0, 4, -3)$.

2. (b) $\overrightarrow{OX} = s\,\overrightarrow{OP} + r\,\overrightarrow{OQ}$.

3. (a) $\pi/4$ and $\pi/3$. (b) The dot product is $c - 4$; (i) $c = 4$, (ii) $c > 4$.

4. $\mathbf{C} = -4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, $\mathbf{D} = 4\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$.

5. (a) -34 by both expansions. (b) (i) $3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$; (ii) $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

6. (a) $\overrightarrow{AB} \times \overrightarrow{AC} = 26\mathbf{i} + 18\mathbf{j} + 6\mathbf{k}$; parallelogram $2\sqrt{259} \approx 32.19$, triangle $\sqrt{259}$. (b) 11. (c) Yes — a zero determinant means the three edge vectors are linearly dependent, so all four points lie in a common plane.

7. (a) $x + 2y - 2z = 4$. (b) $x - y - z = 0$. (c) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. (d) $x + y = 1$.

8. (a) $x = 1 + 2t$, $y = -t$, $z = -1 + 3t$. (b) $x = 2 + t$, $y = -1 - t$, $z = -1 + 2t$. (c) $x = 1 + at$, $y = 1 + bt$, $z = 1 + (a + 2b)t$ for any $(a, b) \neq (0, 0)$. (d) Direction $\langle 3, -4, 4\rangle \times \langle 3, -1, 5\rangle = \langle -16, -3, 9\rangle$: $x = 4 - 16t$, $y = 2 - 3t$, $z = 1 + 9t$.

9. (a) $(4, -1, 4)$. (b) $(0, -1, 0)$. (c) $x - 2y + 4z + 1 = 0$.

Part B

10. (a) $\mathbf{v} = e^t\mathbf{i} - e^{-t}\mathbf{j}$, speed $\sqrt{e^{2t} + e^{-2t}}$, $\mathbf{T} = \mathbf{v}/|\mathbf{v}|$, $\mathbf{a} = e^t\mathbf{i} + e^{-t}\mathbf{j}$. (b) $\mathbf{v} = 2t\mathbf{i} + 3t^2\mathbf{j}$, speed $|t|\sqrt{4 + 9t^2}$, $\mathbf{T} = (2\mathbf{i} + 3t\mathbf{j})/\sqrt{4 + 9t^2}$ for $t > 0$, $\mathbf{a} = 2\mathbf{i} + 6t\mathbf{j}$. (c) $\mathbf{v} = t(-4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k})$, speed $6|t|$, $\mathbf{T} = \frac{1}{3}(-2\mathbf{i} + \mathbf{j} + 2\mathbf{k})$ for $t > 0$, $\mathbf{a} = -4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ (constant).

11. (a) $\mathbf{v}' = (6t + 2)\mathbf{i} + \pi^2\cos(\pi t)\mathbf{j} + e^t\mathbf{k}$; $\mathbf{v}'(2) = 14\mathbf{i} + \pi^2\mathbf{j} + e^2\mathbf{k}$. (b) $(t^3 + t^2)\mathbf{i} - \cos(\pi t)\mathbf{j} + e^t\mathbf{k} + \mathbf{C}$. (c) $2\mathbf{i} + 2\mathbf{j} + (e - 1)\mathbf{k}$.

12. $\mathbf{r}(1) = \mathbf{0}$ and $\mathbf{r}'(1) = \mathbf{i} + \frac{1}{4}\mathbf{j} + \mathbf{k}$, so $x = t$, $y = t/4$, $z = t$.

13. $5\mathbf{i} + 5\mathbf{j} + \frac{5\pi}{4}\mathbf{k}$.

14. (a) No: $t = 1 + 2t$ forces $t = -1$, and then $t^2 = 1 \neq -5$. (b) Yes. Solving $s = 1 + 2t$, $s^2 = 1 + 6t$, $s^3 = 1 + 14t$ gives $s = 1$ ($t = 0$) and $s = 2$ ($t = \frac{1}{2}$); the paths meet at $(1, 1, 1)$ and $(2, 4, 8)$.

15. (a) $|\mathbf{v}| = 3$, $\mathbf{T} = \frac{1}{3}\big(-2\sin t\,\mathbf{i} + 2\cos t\,\mathbf{j} + \sqrt{5}\,\mathbf{k}\big)$, length 3π. (b) $|\mathbf{v}| = 4|\sin(t/2)|$, length 32. (c) $\sqrt{3} + \ln\big(2 + \sqrt{3}\big) \approx 3.049$.

Part C

16. (a) $4/5$ and 3. (b) $\{(x, y) : 0 \leq x^2 + y^2 < 9$ and $x^2 + y^2 \neq 8\}$: the open disk of radius 3 with the circle of radius $2\sqrt{2}$ removed. (c) For $0 \leq c < 4$, the sphere $x^2 + y^2 + z^2 = 16 - c^2$ centred at the origin; for $c = 4$, the origin alone.

17. (a) (i) parallel lines $x + y = 1 - c$, graph a plane; (ii) circles $x^2 + y^2 = c^2$, $c \geq 0$, graph a cone; (iii) circles of radius $\sqrt{c}$, graph a paraboloid opening upward; (iv) circles $x^2 + y^2 = 1 - c$, graph a paraboloid opening downward. (b) Lines through the origin (origin and the line $y = x$ deleted): $y = \frac{k-1}{k+1}x$, i.e. $y = 3x$, $y = -x$, $y = 0$, $y = \frac{1}{3}x$ for $k = -2, 0, 1, 2$.

18. (a) 1. (b) $\frac{1}{4}$ (multiply by $\sqrt{2x - y} + 2$). (c) Does not exist: the limit is 0 along $y = 0$ but 1 along $y = x^2$. (d) The limit is $0 \neq f(0, 0) = 1$, so f is not continuous at the origin.

19. (a) $12x^2 - 6xy^2 + 2$; $-6x^2y$. (b) $3\cos(3x + 2y)$; $2\cos(3x + 2y)$. (c) $\frac{4y}{4x+3y}$; $\ln(4x + 3y) + \frac{3y}{4x+3y}$; at $(2, 1)$: $\frac{4}{11}$ and $\ln 11 + \frac{3}{11}$. (d) $2xz$; $-2z^3$; $x^2 - 6yz^2$. (e) $3x^2yz^3$; $2yz + x^3z^3$; $y^2 + 3x^3yz^2$; at $(1, 2, 4)$: 384, 80, 100.

20. (b) $f_{xy} = ax + 6y$ and $f_{yx} = 2x + 6y$, so $a = 2$ and $f = x^2y + 3xy^2 + C$. (c) $f_{xy} = Ax$ and $f_{yx} = 6x$, so $A = 6$ and $f = x^3 + 3x^2y + 2y^3 + C$.

21. (b) $\frac{\partial z}{\partial x} = \frac{3x^2+6z^3}{59-18xz^2}$, $\frac{\partial z}{\partial y} = \frac{6y^2}{59-18xz^2}$.

22. (a) $z = x + 2y - 2$. (b) $w = -4x + 4y$. (c) $z = 8x + 3y - 8$. (d) $4x - 4y - z = 0$.

23. $\nabla F = \langle 2, 2, 1\rangle$ at $(0, 1, 2)$; tangent plane $2x + 2y + z = 4$, normal line $x = 2t$, $y = 1 + 2t$, $z = 2 + t$.

24. (a) $z_0 z = x_0 x + y_0 y$. (b) The ray is (tx_0, ty_0, tz_0), $t \geq 0$; substituting gives $tz_0 = tz_0$ on the cone and $tz_0^2 = t\big(x_0^2 + y_0^2\big) = tz_0^2$ on the plane. (A cone is tangent to each of its tangent planes along a whole ruling.)

25. (a) $\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}$. (b) $xy^2z^3\Big(\frac{dx}{x} + \frac{2\,dy}{y} + \frac{3\,dz}{z}\Big)$. (c) $\frac{2y\,dx - 2x\,dy}{(x+y)^2}$.

26. (a) $dw = -\frac{yw\,dx + (xw + w^3)dy + 2z\,dz}{xy + 3yw^2}$. (b) $dw = -\frac{u\,du + 2v\,dv}{3w}$.

27. (a) $6t^5$. (b) $-2\sin 2t$. (c) 0 (here $u^2 + v^2 \equiv 4$).

28. $\frac{dw}{dt} = 4t\tan^{-1}t + 1$ by either method.

29. $\nabla T = \langle 4, -3, -2\rangle$ at $(0, 2, 3)$ and $\mathbf{r}'(0) = \langle 1, 0, 1\rangle$, so $dT/dt = 2\,^\circ\mathrm{C/s}$.

30. (a) $w_r = f_x\cos\theta + f_y\sin\theta$ and $w_\theta = r\big(-f_x\sin\theta + f_y\cos\theta\big)$; squaring and adding gives the identity. (b) Differentiate w_r again with respect to r, using the Chain Rule on f_x and f_y and $f_{xy} = f_{yx}$.

31. $w_u = 2u\,w_x + 2v\,w_y$ and $w_v = -2v\,w_x + 2u\,w_y$; the cross terms cancel when the squares are added.

32. (a) $w_x = -\frac{y}{x^2}f'$, $w_y = \frac{1}{x}f'$. (b) $w_x = 2xf'$, $w_y = -2yf'$. (c) $w_x = af'$, $w_y = bf'$.

33. (a) $\nabla f = \langle 10, -20\rangle$, $D_{\mathbf{u}}f = -4$. (b) $\nabla f = \langle 1, 3, 0\rangle$, $D_{\mathbf{u}}f = 3$. (c) $\nabla f = \langle 2, 4, -6\rangle$, $D_{\mathbf{u}}f = 0$. (d) $\nabla f = \langle 4, 8, 12\rangle$, $D_{\mathbf{u}}f = -\frac{4}{3}$.

34. $\nabla f(P) = \langle 4, 2\rangle$. (a) (i) $3\sqrt{2}$; (ii) $\frac{58}{13}$. (b) $\frac{1}{\sqrt{5}}\langle 2, 1\rangle$, with value $|\nabla f| = 2\sqrt{5}$. (c) $-\frac{1}{\sqrt{5}}\langle 2, 1\rangle$, with value $-2\sqrt{5}$. (d) $\pm\frac{1}{\sqrt{5}}\langle 1, -2\rangle$.

35. $\nabla f(-6, 2) = \langle 60 + 2a, 432 + 12a\rangle$ and $D_{\mathbf{u}}f = (924 + 26a)/\sqrt{5} = 216\sqrt{5}$, so $a = 6$.

36. $\nabla F = \langle 7, 3, 2\rangle$, $\nabla G = \langle 6, -1, -2\rangle$, and $\nabla F \times \nabla G = \langle -4, 26, -25\rangle$: $x = 1 - 4t$, $y = 2 + 26t$, $z = 1 - 25t$.

37. (a) $L(x, y) = 1 + 2x + y$ at $(0, 0)$; $L(x, y) = e^4(2x + y - 3)$ at $(1, 2)$. (b) $df = \Big(\frac{1}{3} + \frac{\sqrt{3}\pi}{6}\Big)(0.03) + \frac{\sqrt{3}}{12}(0.04) - \Big(\frac{1}{2} + \frac{\sqrt{3}\pi}{4}\Big)(0.02) = \frac{\sqrt{3}}{300} \approx 0.0058$.

38. (a) $1 + x + 4y + 4xy + 8y^2$. (b) 1.76 (the true value is 1.7858…).

Part D

39. (a) Local minimum -5 at $(-3, 3)$ ($D = 3$, $f_{xx} = 2$). (b) Saddle at $(0, 0)$ ($D = -4$); local maximum $\frac{170}{27}$ at $\big(-\frac{2}{3}, \frac{2}{3}\big)$ ($D = 12$, $f_{xx} = -4$). (c) Saddle at $(0, 0)$; local maxima of

value 2 at $(1,1)$ and $(-1,-1)$. (d) Local maximum $-3-2\ln 2$ at $\left(\frac{1}{2},1\right)$ ($D=8$, $f_{xx}=-8$).

40. Critical points $(0,0)$, $(2,0)$, $(0,2)$, $\left(\frac{2}{3},\frac{2}{3}\right)$; $D=4xy-(2-2x-2y)^2$. The first three are saddle points ($D=-4$); the last is a local maximum with value $\frac{8}{27}$ ($D=\frac{4}{3}$, $f_{xx}=-\frac{4}{3}$). There is no local minimum.

41. The only critical point is $\left(2,\frac{1}{2}\right)$, with $D=1>0$ and $f_{xx}=-\frac{1}{4}<0$: a local maximum of value $-2-\ln 2$. There is no local minimum and no saddle point.

42. Interior critical point $(1,1)$ with $f=-1$. On the boundary the candidates are $f(-1,0)=1$, $f(-1,2)=9$, $f(2,0)=4$, $f(2,2)=0$, $f\left(2,\frac{3}{2}\right)=-\frac{1}{2}$, $f(0,0)=0$. Absolute maximum 9 at $(-1,2)$; absolute minimum -1 at $(1,1)$.

43. (a) Minimise $x^2+y^2+z^2=x^2+xz+z^2+4$; the only critical point is $x=z=0$, giving $(0,\pm2,0)$ at distance 2. (b) $\frac{|2-1-1-2|}{\sqrt{3}}=\frac{2\sqrt{3}}{3}$.

44. (a) $y=\frac{1}{4}x+\frac{17}{12}$. (b) $y=\frac{3}{11}x^2-\frac{4}{11}x$.

45. With width x, depth y, height z: $C=20xz+10yz+30xy$ subject to $xyz=6$. Eliminating z gives $C=\frac{60}{x}+\frac{120}{y}+30xy$, whose only critical point is $x=1$, $y=2$, so $z=3$. Minimum cost \$180, with dimensions 1 m × 2 m × 3 m (each of the three cost terms equals \$60).

46. (a) The equations give $y^2=x^2-2x$, hence $x=0$ or $x=3$; the maximum is $3\sqrt{3}$ at $\left(3,\sqrt{3}\right)$ (and the minimum is $-3\sqrt{3}$ at $\left(3,-\sqrt{3}\right)$). (b) The candidate $x=3$ is excluded, so the maximum occurs on the new boundary $x=2$: the maximum is 4, at $(2,2)$.

47. (a) $\left(-\frac{2}{3},-\frac{1}{3},-\frac{2}{3}\right)$, at distance 4. (b) $\left(\frac{3}{2},2,\frac{5}{2}\right)$.

48. At the maximum $\frac{x^2}{3}=\frac{y^2}{9}=\frac{z^2}{4}=\frac{1}{3}$, so $x=1$, $y=\sqrt{3}$, $z=\frac{2}{\sqrt{3}}$ and $V_{\max}=2$. The constraint surface is closed and bounded and V is continuous, so the extreme values are attained; $V=0$ on the boundary of the first octant, so the interior critical point is the maximum.

49. (a) $(0,0)$ is the only critical point, and $D=-4<0$: a saddle point, $f(0,0)=0$. (b) On $x^2+y^2=1$: maximum 1 at $(\pm1,0)$, minimum -1 at $(0,\pm1)$.

50. (a) On $y=x$ the constraint becomes $2x^2+z^2=4$ and $f=4-x^2$: maximum 4 at $(0,0,\pm2)$, minimum 2 at $\pm\left(\sqrt{2},\sqrt{2},0\right)$. (b) The planes meet where $x+y=20$, $z=20$, so $w=20xy$ with $x+y=20$; the maximum is $w=2000$ at $(10,10,20)$. Along the line $w=20x(20-x)\to-\infty$ in both directions, so the critical point must be a maximum.

51. (a) $T=(2x-y)^2$; on $x=5\cos t$, $y=5\sin t$ this is $25(2\cos t-\sin t)^2$, so the hottest temperature is 125° and the coldest is 0°. (b) $r=2$ cm, $h=4$ cm. (c) $U_{\max}=128$ at $x=8$, $y=14$.

52. The method gives $x=y$, so $(4,4)$ and $(-4,-4)$. On the branch in the first quadrant $f=x+16/x\to+\infty$ as $x\to0^+$ and as $x\to+\infty$, so $(4,4)$ is a *minimum* there ($f=8$) and f has no maximum on $xy=16$. On the third-quadrant branch $(-4,-4)$ gives a maximum, $f=-8$. The constraint set is unbounded, which is exactly what $\nabla f=\lambda\nabla g$ cannot detect.

Part E

53. $\overrightarrow{AB}\times\overrightarrow{AC}=\langle5,3,4\rangle$. (a) $\frac{5\sqrt{2}}{2}$. (b) $5x+3y+4z=13$. (c) $\frac{13}{5\sqrt{2}}=\frac{13\sqrt{2}}{10}$.

54. (a) $\mathbf{v}=t\cos t\,\mathbf{i}+t\sin t\,\mathbf{j}$, $|\mathbf{v}|=t$ for $t\ge0$. (b) $s(t)=t^2/2$; the length is $\pi^2/8$. (c) $\mathbf{T}=\cos t\,\mathbf{i}+\sin t\,\mathbf{j}$. (This curve is the involute of the unit circle.)

55. (a) $\nabla f(P)=\langle4,2,-4\rangle$. (b) $-\frac{2}{3}$. (c) In the direction $\frac{1}{3}\langle2,1,-2\rangle$, at the rate $|\nabla f|=6$. (d) $\mathbf{r}(1)=\langle1,2,-1\rangle=P$ and $\mathbf{r}'(1)=\langle1,1,2\rangle$, so $df/dt=4+2-8=-2$.

56. $\nabla F(P_0)=\langle2,8,12\rangle$. (a) Tangent plane $x+4y+6z=21$; normal line $x=1+t$, $y=2+4t$, $z=2+6t$. (b) $\partial z/\partial x=-x/(3z)=-\frac{1}{6}$ and $\partial z/\partial y=-2y/(3z)=-\frac{2}{3}$, matching $z=\frac{1}{6}(21-x-4y)$. (c) $\nabla F\parallel\langle1,4,6\rangle$ forces $(x,y,z)=\pm(1,2,2)$, so the points are $(1,2,2)$ and $(-1,-2,-2)$.

57. $f_x=0$ and $f_y=0$ give $x^2+y^2=5$ and $xy=2$, so the critical points are $(2,1)$, $(1,2)$, $(-1,-2)$, $(-2,-1)$. Here $D=36\left(x^2-y^2\right)$. Local minimum -28 at $(2,1)$; local maximum 28 at $(-2,-1)$; saddle points at $(1,2)$ and $(-1,-2)$.

58. (a) $\langle1,-2,5\rangle=2\lambda\langle x,y,z\rangle$ gives $(x,y,z)=\pm(1,-2,5)$: maximum 30 at $(1,-2,5)$, minimum -30 at $(-1,2,-5)$. (b) A cube of edge $\frac{4}{\sqrt{3}}$, of volume $\frac{64}{3\sqrt{3}}=\frac{64\sqrt{3}}{9}$.

Problems adapted from *Thomas' Calculus: Early Transcendentals*, 15th edition (Hass, Heil, Weir), §§11.1–11.5, 12.1–12.3, 13.1–13.9; from the MIT 18.02 problem sets distributed with the Lecture Group 1 lesson notes; and from the MA1521 Lecture Group 2 tutorial questions, Lessons 02–12.